

PATRICK TINSLEY

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EDUCATION

University of Notre Dame

PhD in Computer Science and Engineering - GPA: 3.92

Dissertation Title: "Trust, AI, and Synthetic Biometrics" (defended April 2024)

Notre Dame, IN

June 2018 – May 2024

University of Notre Dame

MSc in Applied and Computational Mathematics and Statistics - GPA: 3.88

Specialty in Predictive Analytics (Time Series, Causal Inference, Applied Probability, Linear Models, Recommender Systems)

Notre Dame, IN

June 2017 – May 2018

University of Notre Dame

BSc in Applied and Computational Mathematics and Statistics - GPA: 3.75 (Cum Laude)

Relevant Coursework: Advanced Biostatistical Methods, Neural Networks, Computational Methods, Statistical Methods in Data Mining, Numerical Analysis, Computational Modeling for Neuroscience, Web Applications, C/C++ Programming

Notre Dame, IN

August 2013 – May 2017

PROFESSIONAL EXPERIENCE

AngelEye Health

Manager of Medical Technology

(remote) Nashville, TN

June 2024 – Present

- Trained pytorch-based anomaly detection models on pose estimate data to detect cramped synchronized (CS) movement patterns from video; visualized results with seaborn to aid human assessors when grading movements.
- Developed a real-time alert system to notify nurses when an infant may be at risk of falling from their crib in the neonatal ICU (NICU); the mat-based pressure sensor minimizes power consumption and alert inference time.
- Partnered with Nationwide Children's Hospital to augment video data with pressure sensor data for the 5,000 Babies Project; conducted extensive pattern analysis on pose estimate data from 3500+ videos in order to detect abnormal movements; handcrafted features for evaluating symmetry and cohesion in limb movement for camera and mat data.

University of Notre Dame

Research/Teaching Assistant

Notre Dame, IN

June 2018 – May 2024

- Introduced and quantitatively assessed the phenomenon of identity leakage in synthetic biometric image generation, wherein AI-synthesized images falsely match authentic images seen during training; experiments covered three major biometric modalities: face, iris, and fingerprint.
- Organized and moderated the 2023 edition of the iris liveness detection competition (LivDet-Iris); generated 6,000 distinct synthetic iris images of varying image quality using generative adversarial networks (GANs) to validate and rank algorithmic performance for seven team submissions.
- Designed and performed extensive matching experiments with state-of-the-art face encoders to evaluate latent space clustering tendencies in the context of large-scale, racially-diverse, pose-variant face images.
- Implemented a progressive web app to interactively conduct tests with and visualize results from person re-identification models for in-the-wild CCTV surveillance videos; project deliverable allowed security officers to upload and identify persons of interest in long-play video streams as well as flag false matches and non-matches for later model re-training.

Aunalytics

Machine Learning Intern

South Bend, IN

June 2017 – September 2017

- Benchmarked and predicted training and inference time for 150+ models with varying neural network architecture, loss function, and training data; developed linear model to estimate B2B deliverable turnaround time given model type and training data size.
- Implemented and maintained internal model zoo with multiple formats of production-ready machine learning models; formats included pickle, .RDa, and .pb (TensorFlow).

SKILLS

- **Programming:** Python (PyTorch, TensorFlow, Keras, scikit-learn), SQL, R, Matlab, C/C++, HTML/CSS/JavaScript.
- **Technical:** Tableau, Git, Docker, Singularity, Google Analytics, AWS.
- **Honors & Awards:** 2018 Graduate Student Union Outstanding TA Winner (Data Science).

PUBLICATIONS

First Author

- Tinsley, Patrick. “Trust, AI, and Synthetic Biometrics.” (2024).
- Tinsley, Patrick, Sandip Purnapatra, Mahsa Mitcheff, Aidan Boyd, Colton Crum, Kevin Bowyer, Patrick Flynn, Stephanie Schuckers, Adam Czajka, Meiling Fang, Naser Damer, Xingyu Liu, Caiyong Wang, Xianyun Sun, Zhaohua Chang, Xinyue Li, Guangzhe Zhao, Juan Tapia, Christoph Busch, Carlos Aravena, Daniel Schulz. “Iris liveness detection competition (livdet-iris)—the 2023 edition.” IEEE International Joint Conference on Bio- metrics (IJCB 2023).
- Tinsley, Patrick, Adam Czajka, and Patrick Flynn. “Haven’t I Seen You Before? Assessing Identity Leakage in Synthetic Irises.” IEEE International Joint Conference on Biometrics (IJCB 2022).
- Tinsley, Patrick, Adam Czajka, and Patrick Flynn. “This Face Does Not Exist... But It Might Be Yours! Identity Leakage in Generative Models.” Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2021).

Second Author

- Boyd, Aidan, Patrick Tinsley, Kevin Bowyer, and Adam Czajka. “Training Better Deep Learning Models Using Human Saliency.” arXiv preprint arXiv:2410.16190 (2024).
- Mitcheff, Mahsa, Patrick Tinsley, and Adam Czajka. “Privacy-Safe Iris Presentation Attack Detection.” IEEE International Joint Conference on Biometrics (IJCB 2024).
- Khan, Siamul Karim, Patrick Tinsley, Mahsa Mitcheff, Patrick J. Flynn, Kevin Bowyer, and Adam Czajka. “EyePreserve: Identity-preserving iris synthesis.” arXiv preprint arXiv:2312.12028 (2023).
- Crum, Colton, Patrick Tinsley, Aidan Boyd, Jacob Piland, Christopher Sweet, Timothy Kelley, Kevin Bowyer, and Adam Czajka. “Explain to me: Saliency-based explainability for synthetic face detection models.” IEEE Transactions on Artificial Intelligence (TAI 2023).
- Boyd, Aidan, Patrick Tinsley, Kevin Bowyer, and Adam Czajka. “The Value of AI Guidance in Human Examination of Synthetically-Generated Faces.” Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI 2023).
- Khan, Siamul Karim, Patrick Tinsley, and Adam Czajka. “DeformIrisNet: An Identity-Preserving Model of Iris Texture Deformation.” Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2023).
- Boyd, Aidan, Patrick Tinsley, Kevin Bowyer, and Adam Czajka. “CYBORG: Blending Human Saliency Into the Loss Improves Deep Learning.” Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2023).
- Iloanusi, Ogechukwu, Patrick J. Flynn, and Patrick Tinsley. “Similarities in African Ethnic Faces from the Biometric Recognition Viewpoint.” Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2022).

LEADERSHIP AND SERVICE

South Bend Center for the Homeless (CFH)

South Bend, IN

Volunteer

May 2018 – August 2024

- Performed a lip sync with team “Squad Ghouls” in the 2022 LipSync With Our Stars fundraiser; raised \$26,000.
- Competed and fundraised as a dancer in the 2019 Dancing With Our Stars event; raised \$13,000.
- Organized holiday parties for 50 CFH children and assisted with special projects (painting, landscaping, donations).

University of Notre Dame - Graduate Orientation (GO) Team

Notre Dame, IN

Leader

May 2021 – August 2023

- Directed a 30-member team to plan and execute a week-long orientation for 2000+ incoming graduate students (Master’s and Ph.D.) from all colleges.
- Led the development and delivery of events that fostered community, inclusion, and a smooth transition to the Notre Dame and South Bend communities.